

Space Cleanroom Facility in Slovakia for Heliophysics Missions

ISO 7 cleanroom · vibration · thermal-vacuum · JUICE flight heritage — now open to partners

Š. Mackovjak, J. Baláž, I. Strhárský, D. Fečko, P. Usatiuk

Department of Space Physics · Institute of Experimental Physics, Slovak Academy of Sciences, Košice, Slovakia

25 m²	ISO 5	−100...+160 °C	10^{−7} mbar	5100 N
ISO class 7 clean space	laminar-flow local zone	thermal-vacuum range	vacuum level	sine peak, to 4000 Hz

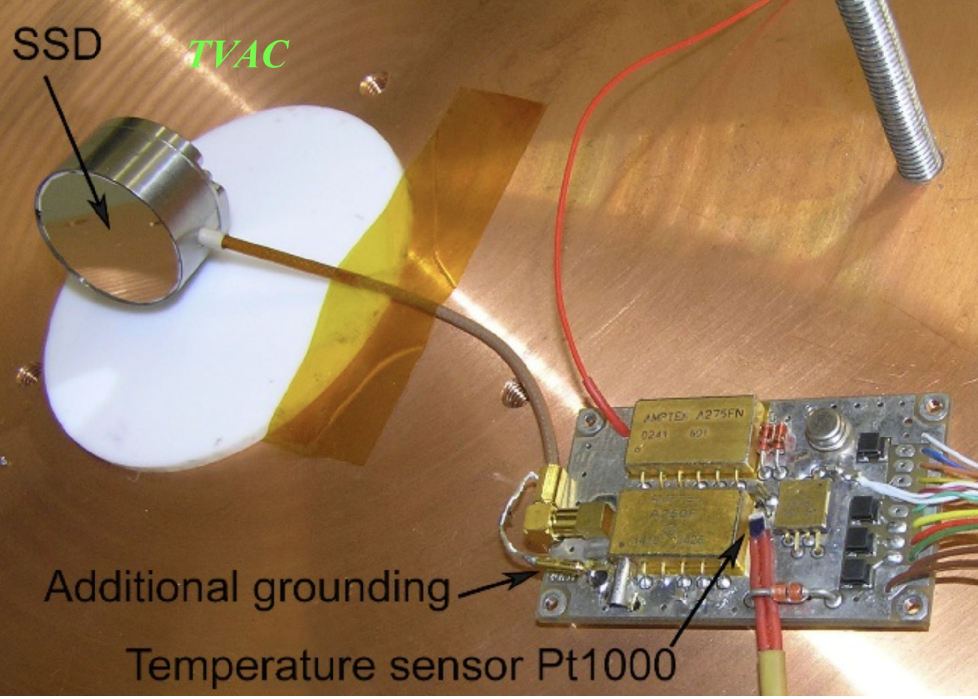
The facility

Slovakia's first space cleanroom provides 25 m² of ISO class 7 clean space with an ISO class 5 laminar-flow zone. Temperature, humidity and particle concentration are controlled continuously, and processes are run to ECSS standards. Personnel enter through an air shower, hardware through a 60 × 60 cm pass-through.



Integration at the ISO 7 cleanroom, with the thermal-vacuum chamber behind.

Flight heritage



Anti-Coincidence Module test configuration inside the thermal-vacuum chamber (JUICE / PEP).

The Anti-Coincidence Module of PEP on ESA's JUICE was tested in this chamber before delivery to IRF Kiruna.

- Collaboration on other ESA missions:
- Rosetta / ESS
 - Double Star / NUADU
 - BepiColombo / PICAM

Assembly, integration and test

Electronics assembly and inspection

- Soldering workstations and analytical balance
- Ersascope X joint inspection system
- Oscilloscopes, power supplies, multimeters, generators

Vibration — Brüel & Kjær LDS V780

- Frequency range up to 4000 Hz
- Excitation force up to 5100 N (sine peak)
- Acceleration up to 1050 m·s^{−2} (sine peak)
- Chronos 2.1-HD high-speed camera

Thermal-vacuum — Vakuu Praha SPACEVAC

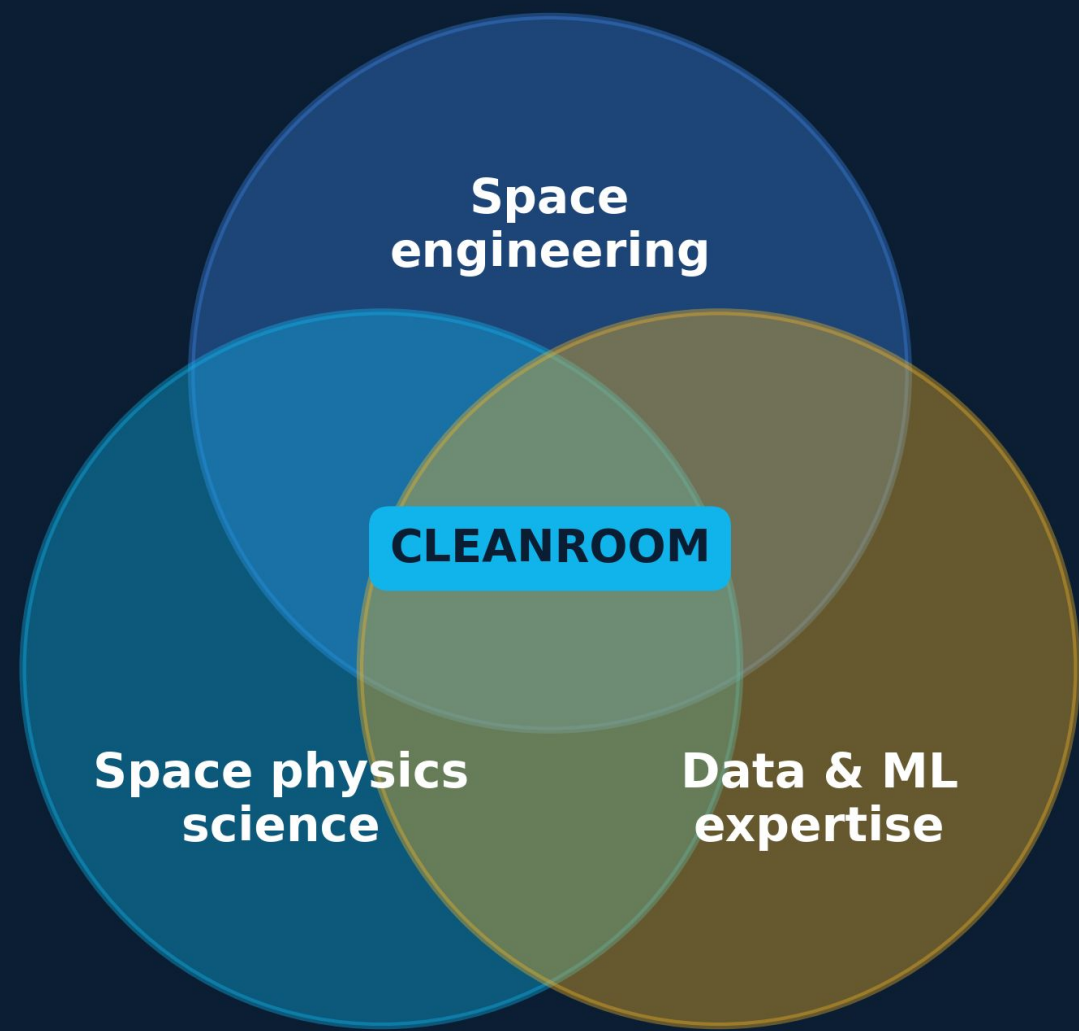
- −100 °C to +160 °C
- Vacuum to 1 × 10^{−7} mbar
- Flight-qualified chamber (JUICE / PEP)

Devices under test are exercised over their flight interfaces.



Test automation and flight interfaces

NI modules: temperature, voltage, digital I/O, relays · SpaceWire, MIL-STD-1553, Ethernet, CAN, RS-485/422



WORK WITH US

Payload electronics for heliophysics missions.

01

Host your instrument

We have capabilities to integrate and test heliophysics payloads, including those of the national STEPHANIK platform.

02

Run an AIT campaign

Vibration and thermal-vacuum qualification to ECSS, with National Instruments test automation.

03

Co-develop on-board processing

Machine learning designed with the instrument, not bolted on after it — as in Vigil-ML and SCSS-Net.

space-lab@saske.sk